

Kousha Ghodsizad

Systems & Infrastructure Engineer

CONTACT

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TECHNICAL SKILLS

Languages & Backend
Go, Python, PHP, Shell, APIs, gRPC, GraphQL, services, workers, CLIs

Containers & Orchestration
Kubernetes, K3s, Docker, Helm

Kubernetes Networking & Storage
Cilium (Kubernetes networking), Calico (network and firewall controls in existing infrastructure), Flannel (pod networking in personal infrastructure), Longhorn (storage in personal infrastructure)

Delivery & Workflows
Argo CD, Argo Rollouts, Argo Workflows, Argo Events, Temporal, GitLab CI, Jenkins, Harbor, Nexus

Infrastructure & Automation
AWS, private cloud, OpenStack, Terraform, Puppet, HashiCorp Vault, HashiCorp Consul

Data & Messaging
MySQL, PostgreSQL, Redis, RabbitMQ, Kafka, NATS, Cassandra, ClickHouse, Elasticsearch, MongoDB

Observability & Reliability
Prometheus, Grafana, metrics, logging, operational dashboards, readiness checks, rollout safety

Architecture
System design, reliability engineering, automation strategy, modular architecture, domain boundaries, technical review, engineering standards

SUMMARY

Systems & Infrastructure / Platform Engineer with a backend engineering background. I build and operate production platforms, from Kubernetes infrastructure and CI/CD workflows to the developer tools engineering teams use to build and run their services. My work spans infrastructure automation, observability, reliability, and distributed systems, with a focus on secure, resilient, and maintainable systems.

I take ownership from design and implementation through production operation and ongoing improvement. I enjoy solving complex technical problems, making architectural decisions, and building tools that help engineering teams work independently. I care about engineering quality and systems that remain reliable as they scale.

CORE EXPERTISE

Platform Engineering & Developer Tooling: Go-based platform tools and CLIs, plugin management, secure execution, reusable automation, and self-service workflows.

Kubernetes & Application Delivery: Kubernetes and K3s infrastructure, Helm, GitOps with Argo CD, progressive delivery with Argo Rollouts, and CI/CD with GitLab CI and Jenkins.

Infrastructure as Code & Secrets: Infrastructure provisioning with Terraform, configuration management with Puppet, secrets management with HashiCorp Vault, and service discovery with HashiCorp Consul.

High Availability & Reliability: Multi-data-centre architectures, request routing, workflow recovery and failover, deployment safety, and production reliability.

Observability & Production Operations: Prometheus, Grafana, metrics, logging, dashboards, post-deployment monitoring, and production issue investigation.

Backend Systems & Architecture: Go services, APIs, workers, messaging, distributed systems, system design, service boundaries, technical reviews, and engineering standards.

EXPERIENCE

Adyen

Jul 2023 — Present

Infrastructure Engineer — CI/CD & Platform · Amsterdam, Netherlands

- Designed and built **`adyencli` in Go**, the CLI for Adyen's plugin management platform. Engineering teams can develop plugins using Adyen-provided SDKs in four programming languages, submit them for building and distribution through Adyen's package-management infrastructure, and manage their lifecycle and controlled execution on Linux.
- Implemented access controls and secure, controlled plugin execution. Built platform services within Adyen's existing infrastructure, which included Kubernetes, Helm, Argo CD, Cilium networking, Consul for service discovery, Vault for secrets management, and Calico-based network and firewall controls between data centres.
- Designed the backend architecture of a separate highly available system distributed across four geographically separated data centres. Requests were routed to suitable, responsive services, and the architecture was built to handle heavy workloads and remain operational when one or more data centres became unavailable.
- Built a highly available, multi-cluster execution solution for Argo Workflows across Kubernetes clusters and data centres. Every workflow was available across the participating clusters, while one cluster owned and executed its steps at a time.

- Coordinated execution ownership per workflow, with Kubernetes Leader Election as part of the broader execution architecture. If the owning cluster failed or came under resource pressure, another available cluster with sufficient capacity could take over that workflow's execution.
- Enabled Argo Workflows node-status offloading as part of the recovery design. Workflow node and step state was persisted through the database-backed setup. When a node failure caused a step to be marked `Failed`, the new execution owner could use the recorded state and retry handling to rerun that step and continue from that point rather than restarting the entire workflow.
- Contributed to a modular, self-service automation platform built in Python. Teams define repository-specific automation through Gherkin scenarios that invoke reusable modules. The platform uses NATS and provides detailed Prometheus and Grafana monitoring down to individual merge requests and their pipeline executions.
- Worked with Jenkins and GitLab CI for pipeline automation, Argo Rollouts for progressive delivery, Argo Events for event-driven automation, Temporal for workflow orchestration, and Terraform and Puppet for infrastructure provisioning and configuration management.

Go · Python · Gherkin · Kubernetes · Helm · Argo CD · Argo Workflows · Argo Events · Argo Rollouts · Temporal · Jenkins · GitLab CI · Terraform · Puppet · HashiCorp Vault · HashiCorp Consul · Cilium · Calico · NATS · Redis · Prometheus · Grafana · Linux

Studocu

Jul 2022 — Jul 2023

Senior Software Engineer · Amsterdam, Netherlands

- Improved backend and platform-facing functionality related to website performance, crawler accessibility, and production reliability.
- Worked with operations and platform teams to improve monitoring across local, staging, and production environments.
- Built CI/CD checks for SEO and performance testing, making release quality easier to validate consistently.
- Investigated application and database bottlenecks and translated findings into practical engineering improvements.

AWS · S3 · SNS · SQS · RDS · PHP · Laravel · Go · MySQL · Elasticsearch · Redis · TDD

KolayIK

May 2021 — May 2022

Senior Software Engineer / Backend Team Lead · Istanbul, Turkey

- Led backend engineering across HR systems, expense management, data import/export, and queue-based services.
- Designed and brought up K3s infrastructure from scratch for a multi-data-centre, high-availability environment, covering cluster deployment and the supporting infrastructure required to operate it.
- Applied modular architecture, domain boundaries, test-driven development, messaging, and AWS services to improve maintainability and operational behaviour.
- Built automation for failure reporting and operational feedback, including queue handling and Slack-integrated reporting.
- Supported delivery through technical reviews, design decisions, mentoring, and practical engineering standards.

Wingie Enuygun Group

May 2020 — Apr 2021

Senior Software Engineer / Team Lead · Istanbul, Turkey

- Led hotel-provider integrations and backend services for provider aggregation, search, caching, and data pipelines.
- Designed Go/gRPC services to coordinate concurrent provider requests and standardise hotel data from external systems.
- Improved reliability through testing, Elasticsearch tuning, cache-rate management, and database cluster migration work.
- Guided technical reviews, testing workshops, and delivery decisions that balanced reliability with business needs.

Hexagonal Architecture · Microservices · AWS · PHP · Go · MySQL · MongoDB · Symfony · DevOps · TDD

Insider

Jun 2019 — Jun 2020

Senior Software Engineer · Istanbul, Turkey

- Refactored newsletter and email-delivery systems to establish clearer service boundaries and improve maintainability.
- Worked on Go services, messaging, security remediation, and data migration across high-volume marketing data systems.
- Improved service design through architecture reviews, repository-service patterns, and clearer data flows.
- Built supporting services for A/B testing, spam detection, and queue-based processing.

RabbitMQ · AWS · PHP · Go · MySQL · MongoDB · Redis · Design Patterns · TDD

Bamilo

Aug 2015 — Apr 2019

Senior Software Engineer / Head of Development · Tehran, Iran

- Built and led engineering work across seller tooling, warehouse operations, delivery tracking, inventory services, and internal platforms.
- Owned backend architecture, service design, team growth, and technical reviews for marketplace operations systems.
- Designed data migration pipelines and improved services supporting inventory and seller-centre workflows.
- Turned operational needs into maintainable systems for logistics, warehouse, seller, and marketplace teams.

RabbitMQ · Kubernetes · GraphQL · AWS · PHP · Laravel · Go · MySQL · MongoDB · Redis · Cassandra · Kafka

Sepehr Mahan Tech

Aug 2012 — Jul 2015

Software Engineer / Senior Software Engineer · Tehran, Iran

- Developed suggestion, task-management, meeting-management, form-builder, workflow, social-network, and mobile API systems.
- Built backend services and application foundations focused on process automation, maintainability, and practical delivery for private- and public-sector clients.
- Developed the backend and product-engineering foundations that supported later work in platform engineering, system architecture, and technical leadership.

Linux · PHP · JavaScript · Python · Node.js · SQL · MySQL · Redis · MongoDB · RabbitMQ · Design Patterns

SELECTED PROJECTS

Pepper.com Platform Work

Scalable image processing and marketplace microservices · Remote / Berlin

- Built a Go-based image-processing service capable of processing around **5,000 new images per minute**, generating multiple resized variants for delivery across the platform.
- Implemented deterministic request routing for image-resizing workloads. Cloudflare middleware derived a routing value from each image ID and the configured scaling limit, added it to a request header, and enabled the load-balancing layer to route requests to Kubernetes pods based on that value. This kept requests for a given image consistently routed to one service under the same routing configuration.
- Prevented concurrent duplicate resize requests for the same image from triggering redundant processing. A singleflight-style request-coalescing mechanism allowed overlapping requests to share work already in progress.
- Developed GraphQL flows for deal submission, product listings, and product details, and contributed to image-service performance, near-cache behaviour, load balancing, and microservice security reviews.

Go · Kubernetes · Cloudflare · GraphQL · AWS · SQS · SNS · RDS · S3 · ELB · Elasticsearch · Redis · TDD

EDUCATION

BSc Computer Software Engineering · Islamic Azad University · 2011 — 2015

ASc Computer Programming · Islamic Azad University · 2008 — 2011

RECOMMENDATIONS

Nikola Barac — Director of Engineering at Atolls (former Global Savings Group)
“Kousha worked on pepper.com features as a senior PHP developer and took the ownership of one of our most important microservices written in Go. He demonstrated passion for tackling challenging problems, staying off hours until they got solved and following up the next days. While working on his daily tasks he was often paying attention to security issues that many developers often neglect. Kousha was a valuable team member and I would be very happy to work with him again!”

Morteza Poussaneh — Applied AI Engineer · Ex-CTO
“I am delighted to recommend Kousha as a senior software developer. Having worked closely with him in three different companies, I can confidently say that he is an exceptional professional. Kousha's passion for technology is evident in his eagerness to learn and his ability to effortlessly work with multiple programming languages. He is known for getting things done, even in challenging and stressful situations, and his calm demeanor and problem-solving skills are truly admirable. Kousha's dedication, expertise, and exceptional work ethic make him an invaluable asset to any organization.”

Ryan Klann — VP of Engineering, Pepper.com
“Kousha is a developer who has very strong technical skills and has shown to be able to take responsibility for a project and be relied upon. If there is something technical that he doesn't understand he will jump right in and learn what he needs in order to solve the problem. Its been a pleasure working with you.”

Gizem Sevinç Erdur — Former Founder & CTO, HR Tech
“Kousha is a talented software developer that I thoroughly enjoy working with. Thanks to its extensive knowledge in backend technologies, it provides effective and fast solutions to many problems. Highly recommended if you are working with PHP, Phyton or Go.”

References available on request.